

Cell Design Specification - VBF-T5R1-DS-01

VBF-T5R1 delivery package - generated 2026-08-25

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Case: t5_r1 - next-generation flagship vehicle battery.

Contract (log.jsonl entry 0, verbatim): energy density ≥ 500.94 Wh/kg; 4C fast charge without lithium plating; maximum temperature ≤ 60 °C (= 333.15 K, mechanical conversion).

Verdict: ACHIEVED - design R7B, DFN-validated under both Chen2020 and OKane2022 (dual judgment).

Generation date: 2026-08-25. All values below are mechanically taken from parameter-set dumps, `bda run-pyamm` outputs, `bda calc-energy` output, or `bda log-evaluate` entries (source annotated per line). Missing items are marked "Not provided".

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite(-SiOx) pouch cell	`--base Chen2020` (recorded default) + `--base OKane2022` cross-validation; OKane2022 negative contains SiOx + native plating/cracking physics
Nominal capacity	5.0 Ah	`Nominal cell capacity [A.h]` (both parameter sets)
Simulation-verified 1C capacity	6.2104 Ah (Chen2020) / 6.2086 Ah (OKane2022), DFN	`cell/r7b_{chen,okane}_1c_dfn.json` via `calc-energy` capacity_ah
Voltage window	2.5 ? 4.2 V	`Lower/Upper voltage cut-off [V]` (both sets)
Cell dimensions (layer stack)	0.065 m ? 1.58 m ? 203.4 ?m	`Electrode height/width [m]` (both sets); thickness = ? layer thicknesses (calc-energy)
Shell/housing thickness	Not provided (no parameter)	parameter set
Electrolyte formulation	EC-based liquid + 1 M LiPF6 (`EC initial concentration in electrolyte` = 4541 mol/m3, `Initial concentration in electrolyte` = 1000 mol/m3) with design-target transport overrides: ? = 2.0 S/m, D_e = 4.5?10^10 m2/s, t? = 0.45	parameter set (base values); overrides from `bridge/r7b_negR261.json`, flagged as literature-upper-bound estimates in proposals R5C/R5D/R6D
Cation transference number	0.45 (design override)	`bridge/r7b_negR261.json`

2. Electrode and separator

Layer	Thickness	Porosity	Material / note	Source
Positive electrode	75.6 ?m	0.335 (set default, not overridden)	NMC811 composite, density 3262 kg/m3	`bridge/r7b_negR261.json` + set dump

Layer	Thickness	Porosity	Material / note	Source
Negative electrode	105.8 ?m	0.45 (override)	graphite(-SiOx) composite, density 1657 kg/m3	same
Separator	8 ?m	0.55 (override)	polyolefin-class, density 397 kg/m3 (set value)	same
Positive current collector	8 ?m Al	-	density 2700 kg/m3	same
Negative current collector	6 ?m Cu	-	density 8960 kg/m3	same
Particle radius (pos/neg)	2.61 ?m / 2.61 ?m	-	high-power small-grain class; negative was the decisive plating lever (R7)	`bridge/r7b_negR261.json`
N/P ratio	1.40	-	= negative/positive thickness ratio 105.8/75.6 (caliber used in proposals R4C?R6B; the parameter set defines fully-charged initial concentrations but has no 0%-SOC stoichiometry keys, so the capacity-density form cannot be computed from set keys alone). Empirically consistent: 1C discharge is anode-limited (R4C: capacity jumped 5.73 -> 6.42 Ah when anode grew 85.2 -> 110 ?m)	propose entries + R4C evaluate
Thermal management	h = 60 W/(m2*K) (design override; set default 10)	-	immersion-class liquid cooling requirement, cooling area 0.00531 m2	`bridge/r7b_negR261.json`; `Cell cooling surface area [m2]`

3. Process design parameters

Parameter	Formula	Value	Source
Positive areal density	$L \cdot (1??) \cdot ?$	$75.6e-6 \cdot (1?0.335) \cdot 3262 = 163.99 \text{ g/m}^2$	mechanical from ?2
Negative areal density	$L \cdot (1??) \cdot ?$	$105.8e-6 \cdot (1?0.45) \cdot 1657 = 96.42 \text{ g/m}^2$	mechanical from ?2
Positive compaction density	$?? \cdot (1??) / 1000$	$3262 \cdot 0.665 / 1000 = 2.169 \text{ g/cm}^3$	mechanical from ?2
Negative compaction density	$?? \cdot (1??) / 1000$	$1657 \cdot 0.55 / 1000 = 0.911 \text{ g/cm}^3$	mechanical from ?2
Electrolyte fill amount	pore volume ? 1.2 g/cm3 (literature density, annotated)	pores = $(75.6e-6?0.335 + 105.8e-6?0.45 + 8e-6?0.55) \cdot 0.1027 \text{ m}^2 = 7.943 \text{ cm}^3 \rightarrow 9.53 \text{ g}$	mechanical; electrolyte density 1.2 g/cm3 = literature default (parameter set lacks the key)
Formation recommendation	-	0.1C CC to 4.2 V, 25 ?C, 2 cycles	design recommended value; actual production-line value requires tuning

4. Mass breakdown (contract caliber: electrolyte excluded)

Area = $0.065 \cdot 1.58 = 0.1027 \text{ m}^2$. Per-layer kg/m2 from `calc-energy` output `layer\_kg\_m2`.

Layer	kg/m2	g/cell	Share
Positive electrode	0.163994	16.842	48.6%
Negative electrode	0.096421	9.902	28.6%
Positive CC (Al 8 ?m)	0.021600	2.218	6.4%
Negative CC (Cu 6 ?m)	0.053760	5.521	15.9%
Separator (8 ?m)	0.001429	0.147	0.4%
Total	0.337204	34.631	100%

Source: `cell/r7b\_chen\_energy.json:layer\_kg\_m2` ? `area\_m2` (mechanical). Electrolyte (9.53 g) excluded from contract mass - `electrolyte\_included: false` annotated in output.

5. Performance verification

Metric	Threshold (entry 0)	Chen2020 (DFN)	OKane2022 (DFN)	Verdict
Energy density	>= 500.94 Wh/kg	642.098 Wh/kg	641.118 Wh/kg	PASS pass (both)
Rated energy	-	22.236 Wh	22.202 Wh	report
Volumetric ED (contract caliber)	-	1064.49 Wh/L	1062.87 Wh/L	report
1C capacity	-	6.210 Ah	6.209 Ah	report
4C charge T_max (45 ?C ambient)	<= 333.15 K	326.829 K (53.7 ?C)	328.189 K (55.0 ?C)	PASS pass (both)
4C plating	false	anode surface potential min +0.01826 V (>0 -> no plating)	min +0.01798 V (>0 -> no plating)	PASS pass (both)
Midpoint voltage	-	3.798 V	3.591 V	report
DC resistance (calc-energy caliber)	-	3.50 m?	17.93 m?	report
4C charge acceptance (physical)	-	4.73 Ah	5.00 Ah	report (see caveat below)

Sources: `cell/r7b\_{chen,okane}\_energy.json`, `cell/r7b\_{chen,okane}\_4c\_dfn.json`, mechanically judged by `bda log-evaluate` round 7 (verdict=pass, checked=3).

Caveat: 4C-output `capacity\_ah` is under-reported 5? by a library bug (`t?C\_rate/3600` missing the 5 Ah nominal factor); physical charge acceptance computed separately as charge-segment duration ? 20 A ? 3600. Does not affect the judgments above (plated from `anode\_potential\_v`, T_max from lumped thermal).

6. Design notes (what changed vs baseline and why - cited from the audit log)

- Thin collectors (Al 16->8 ?m, Cu 12->6 ?m) + slim cathode 75.6 ?m + thick anode 105.8 ?m: ED 400.8 -> 630 Wh/kg class (R1 ceiling assessment -> R3 -> R6; N/P 1.40 unlocks anode-limited 1C capacity, R4C).
- Electrolyte transport overrides ? 2.0 S/m, D_e 4.5e-10 m2/s, t? 0.45, plus ?_neg 0.45 and separator 8 ?m: closed most of the plating gap (?0.13 V -> ?0.016 V, R4?R6) but saturated - literature-upper-bound estimates, flagged.
- Negative particle radius 5.86 -> 2.61 ?m: the decisive fix (R7, after three-strike questioning localized the dip to an end-of-charge solid-phase diffusion overpotential, ? = R2/Ds ~ 1040 s vs ~800 s charge). Anode surface potential crossed positive: +18 mV margin under both systems.
- h = 60 W/(m2*K) immersion-class cooling: T_max <= 60 ?C requirement (R5).
- Full per-round rationale: log.jsonl propose/evaluate entries R1?R7; plan + three-strike update in design_plan.md.